BCG Virtual Internship - Executive Summary

Project Objective

Develop a machine learning model to identify customers likely to churn so that retention actions can be taken proactively.

Key Findings

- The dataset was imbalanced, requiring careful evaluation.
- Margin-related variables showed the strongest (although weak) relationship with churn.
- Feature engineering included an off-peak price difference feature together with additional consumption and contract-based features.

Model Performance

Random Forest Classifier Results:
Accuracy: **91.0%**
Precision: **88.9%**
Recall: **8.5%**
F1-Score: **0.154**
ROC-AUC: **0.70**

Business Impact

The model can support early identification of customers at risk of churn. Although overall accuracy is high, recall is low, indicating additional feature engineering and richer customer behaviour data are needed before production deployment.

Recommendations

- Continue feature engineering.
- Explore XGBoost/LightGBM and hyperparameter tuning.
- Collect richer behavioural features.
- Use the model to support targeted retention campaigns.